FOUR EQUIVALENT CHARACTERIZATIONS OF RESOLVABILITY FOR 2-DIMENSIONAL INTEGRAL NOETHERIAN ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Note-card: the source marks '(some details omitted)' for the regular alteration property over the henselian base; the check (properness and generically-finite persistence under the base change) is carried as an explicit core obligation per the standing note-card precedent.

Editorial note. The (4) -> (3) step cites lemma-equivalence-sequence-normalized-blowups, which is NOT carded (dropped in judgment as a zero-content cross-reference to the blowup-sequence proof) and is declared external.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 1137 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Finiteness of the normalization after a regular alteration. lemma-regular-alteration-implies (carded L26835). Used in (1) implies (4) to make Y^nu -> Y finite.
- Normalizations are normal. Morphisms of Spaces, lemma normalization-normal (cross-chapter, named). Identifies the factorization X -> Y^nu.
- Finiteness in codimension 1. Spaces over Fields, lemma finite-in-codim-1 (cross-chapter, named). Makes the factorization finite over the open V of codimension at most 1 points.
- Normal local rings of dimension at most 2 are Cohen-Macaulay. Algebra, lemmas CM-over-regular-flat and criterion-normal (cross-chapter, named). Gives flatness of the finite morphism over V and regularity of V.
- Blowup sequences under normalized blowups. lemma-equivalence-sequence-normalized-blowups (same chapter, NOT carded, declared external). Assembles the per-neighbourhood normalized blowup sequences into a global one in (4) implies (3).
- Resolution of surfaces for schemes. Resolution of Surfaces, theorem resolve (cross-chapter, named). Provides the normalized blowup sequences resolving the henselian local pieces.

Source reference. Original reference 00R4: focused-seeds/geometry/S9-spres-resolve/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 031S: focused-seeds/geometry/S9-spres-resolve/reftags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07NG: focused-seeds/geometry/S9-spres-resolve/reftags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BHL: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 708–708. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BHT: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 935–935. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BHU: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 971–971. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BHY: focused-seeds/geometry/S9-spres-resolve/reftags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 0BH9: sources/stacks-spaces-resolve/pinned-original-source.tex, source lines 391–391. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BB4: focused-seeds/geometry/S9-spres-resolve/reftags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0AD2: focused-seeds/geometry/S9-spres-resolve/reftags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Theorem 0.1. *Let S be a scheme. Let Y be a two dimensional integral Noetherian algebraic space over S. The following are equivalent*

1. *there exists an alteration $X \to Y$ with X regular,*
2. *there exists a resolution of singularities of Y,*
3. *Y has a resolution of singularities by normalized blowups,*
4. *the normalization $Y^\nu \to Y$ is finite, Y^ν has finitely many singular points $y_1, \ldots, y_m \in |Y|$, and for each i the completion of the henselian local ring $\mathcal{O}^h_{Y^\nu, y_i}$ is normal.*

Proof. The implications (3) $\Rightarrow$ (2) $\Rightarrow$ (1) are immediate.

Let $X \to Y$ be an alteration with X regular. Then $Y^\nu \to Y$ is finite by Lemma 0BHT. Consider the factorization $f : X \to Y^\nu$ from Morphisms of Spaces, Lemma 0BB4. The morphism f is finite over an open $V \subset Y^\nu$ containing every point of codimension ≤ 1 in Y^ν by Spaces over Fields, Lemma 0AD2. Then f is flat over V by Algebra, Lemma 00R4 and the fact that a normal local ring of dimension ≤ 2 is Cohen-Macaulay by Serre's criterion (Algebra, Lemma 031S). Then V is regular by Algebra, Lemma 07NG. As Y^ν is Noetherian we conclude that $Y^\nu \setminus V = \{y_1, \ldots, y_m\}$ is finite. For each i let $\mathcal{O}^h_{Y^\nu, y_i}$ be the henselian local ring. Then $X \times_Y \mathrm{Spec}(\mathcal{O}^h_{Y^\nu, y_i})$ is a regular alteration of $\mathrm{Spec}(\mathcal{O}^h_{Y^\nu, y_i})$ (some details omitted). By Lemma 0BHU the completion of $\mathcal{O}^h_{Y^\nu, y_i}$ is normal. In this way we see that (1) $\Rightarrow$ (4).

Assume (4). We have to prove (3). We may immediately replace Y by its normalization. Let $y_1, \ldots, y_m \in |Y|$ be the singular points. Choose a collection of elementary étale neighbourhoods $(V_i, v_i) \to (Y, y_i)$ as in Section 0BH9. For each i the henselian local ring $\mathcal{O}^h_{Y^\nu, y_i}$ is the henselization of $\mathcal{O}_{V_i, v_i}$. Hence these rings have isomorphic completions. Thus by the result for schemes (Resolution of Surfaces, Theorem 0BHY) we see that there exist finite sequences of normalized blowups

$$X_{i,n_i} \to X_{i,n_i-1} \to \ldots \to V_i$$

blowing up only in points lying over v_i such that X_{i,n_i} is regular. By Lemma 0BHL there is a sequence of normalized blowing ups

$$X_n \to X_{n-1} \to \ldots \to X_1 \to Y$$

and of course X_n is regular too (look at the local rings). This completes the proof. $\square$